TingWei (Wade) Ou

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Education

- **National Yang Ming Chiao Tung University (NYCU)** *September 2023 - February 2027*
  M.S. in Robotics Hsinchu, Taiwan
  - GPA: 3.9/4.3

- **National Yang Ming Chiao Tung University (NYCU)** *September 2019 - July 2023*
  B.S. in Mechanical Engineering Hsinchu, Taiwan
  - GPA: 3.85/4.3

Research Interests

Resource-Efficient Perception, Visual SLAM, 3D Mapping for Mobile Robots and Autonomous Driving

Research Experience

- **Industrial Technology Research Institute, MMSL, Mapping Team** *August 2026 – February 2027*
  Research Intern, supervised by Prof. Chieh-Chih Wang and Dr. Victor Lu Hsinchu, Taiwan
  - **Project: Deep Point Cloud Registration for HD Mapping**
    - Integrating deep point cloud registration into a production HD mapping pipeline to improve robustness over classical scan matching in low-overlap and geometrically degenerate scenes.

- **Human and Machine Lab, NYCU** *September 2023 – February 2027*
  Graduate Research Student, supervised by Prof. Kuu-Young Young Hsinchu, Taiwan
  - **Project: Descriptor Enhancement for Visual SLAM**
    - Addressed unstable data association in feature-based visual SLAM under sparse features and lighting variation by proposing an open-source, plug-and-play descriptor enhancement module that enriches descriptors with spatial context, requiring no modification to existing SLAM pipelines.
    - Outperformed prior descriptor enhancement methods on HPatches matching accuracy (19.0% vs. 10.2% relative gain on ORB), and reduced trajectory error across stereo, stereo-inertial, vision-LiDAR, and multi-camera SLAM systems on public benchmarks.
    - Enabled deployment on resource-constrained robots, sustaining real-time performance (30 FPS on an RTX 4090, 20 FPS on a Jetson Orin NX) with 11× lower GPU memory (544 MiB vs. 6130 MiB) than a representative learned front-end SLAM system.
  - **Project: Autonomous UV Disinfection Robot**
    - Addressed the inefficiency of uniform UV disinfection, which spends equal time on low-risk surfaces, by proposing a hotspot-aware strategy that allocates UV dosage by contact risk, targeting high-touch surfaces where contamination concentrates.
    - Formulated manipulator speed optimization as a linear program over a physics-based irradiance model (within 10% of measured UV-C dose), guaranteeing each surface point its risk-specified minimum dosage.
    - Demonstrated in hospital wards and clinics that the strategy meets the same dosage requirements as full-coverage baselines while reducing execution time by ∼31%.

- **Networked Control Robotics Lab, NYCU** *March 2020 – October 2022*
  Undergraduate Research Student, supervised by Prof. Teng-Hu Cheng Hsinchu, Taiwan
  - **Project: Autonomous Mobile Robot for TDK Robocon**
    - Built a mecanum-wheeled autonomous robot from scratch as part of a student team, completing four sequential manipulation tasks (ball grasping, hoop recognition and shooting, bowling) fully autonomously within a 4-minute limit; awarded Honorable Mention.
    - Designed the robot chassis and its control architecture spanning an onboard PC and Arduino, and fused LiDAR odometry with triangulation to maintain reliable localization despite mecanum-wheel slippage and dynamic bystanders.

Publications

J = Journal, C = Conference

[J] **Ting-Wei Ou**, Huang-Ting Lin, and Kuu-Young Young. Desc++: Efficient Descriptor Enhancement for Data Association in Existing Visual SLAM Systems. *IEEE Access*, 2026. [DOI: 10.1109/ACCESS.2026.3743499]

[C] **Ting-Wei Ou**, Jia-Hao Jiang, Guan-Lin Huang, and Kuu-Young Young. Development of an Autonomous Mobile Robotic System for Efficient and Precise Disinfection. In *Proceedings of the IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 3907–3912, 2025. [DOI: 10.1109/SMC58881.2025.11342560]

Conference Presentations

O = Oral, P = Late-Breaking Results Poster, A = One-Page Abstract

[P] Desc++: Efficient Descriptor Enhancement for Data Association in Existing Visual SLAM Systems. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Pittsburgh, USA, 2026.

[O] Development of an Autonomous Mobile Robotic System for Efficient and Precise Disinfection. *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Vienna, Austria, 2025.

[A] Application of BERT-based Language Model for Visual SLAM in Feature Selection. *International Conference on Advanced Robotics and Intelligent Systems (ARIS)*, Taipei, Taiwan, 2024.

Open Source Projects

- **Desc++: Descriptor Enhancement for Visual SLAM** [GitHub]
  - Released code and pretrained models for ORB, SIFT, SuperPoint, and ALIKE descriptors, with drop-in integration for ORB-SLAM2/3, RGB-L, and MAVIS-SLAM.
- **Efficient Deployment of Learned Feature Descriptors for Visual SLAM** [GitHub]
  - Quantized and integrated FeatureBooster into ORB-SLAM2, running in real time on a laptop RTX 2050 with over 50% smaller model size.
- **ROS-Based Implementation of Multi-Camera-Inertial SLAM** [GitHub]
  - Developed a ROS wrapper for OpenMAVIS that runs directly on live rosbag streams, removing the offline data-conversion step required by the original release.

Work Experience

- **Hong Lang Technology Co., Ltd.** *January 2023 – August 2024*
  Robotics Engineer Remote
  - Led a 4-member team developing a robotic plasma-treatment system to replace dusty, skill-dependent manual surface roughening in footwear manufacturing.
  - Designed geometry-based trajectory generation that automatically converts the reconstructed shoe surface into plasma-processing paths, adapting to different shoe models without manual robot teaching and reducing per-shoe processing time from 3–5 minutes (manual) to ∼30 seconds in industrial demonstrations.

Teaching and Mentoring

- **Teaching Assistant, Robotics (535311, Graduate Course), ECE, NYCU** *February 2025 – June 2025*
  - Supervised course projects on forward/inverse kinematics and simulation of a Stanford Arm.
- **Undergraduate Research Mentor, NYCU – Visual SLAM in Dynamic Environments** *September 2024 – September 2025*
  - Guided an undergraduate student in integrating YOLOv8-based dynamic object detection with ORB-SLAM2 to improve localization robustness in dynamic scenes.

Leadership Experience

- **Team Leader, NYCU iTron Robotics Team** *March 2020 – March 2021*
  - Led a student robotics team of ∼15 members, covering recruitment, technical training, and project management across nationwide competitions.

Honors and Awards

- 2024–2026 Graduate Research Scholarship, NYCU Robotics
- 2025 NSTC Graduate Student International Conference Travel Grant
- 2023 Master's Program Entrance Scholarship, NYCU Robotics
- 2021 Honorable Mention Award, 26th TDK Robocon (Autonomous Robotics Group)

Technical Skills

- **Programming:** Python, C++, C
- **Robotics:** ROS, Visual SLAM, Visual-Inertial SLAM, Navigation, Localization, Sensor Fusion
- **Machine Learning & Computer Vision:** PyTorch, OpenCV, TensorRT
- **Development Tools:** Linux, Git, Docker, CMake, Eigen, PCL
- **Simulation & Hardware Design:** Gazebo, SolidWorks, Fusion 360, AutoCAD
- **Languages:** Chinese (Native), English (TOEFL iBT: 5.0/6.0; 100 on legacy 0–120 scale)